

Insuficiência Hepática Aguda (IHA)

Professor (a):

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Definição

Hepatite

***Transaminases > 2 a 3x valores
referência***

Hepatite Aguda Grave

***Hepatite com icterícia +
alargamento INR > 1,5***



Definição

IHA

***Icterícia + Encefalopatia hepática +
RNI > 1.5 + quadro agudo (< 6 meses)***

< 7d



Hiperaguda

7d - 12 sem



Subaguda

12 - 24 sem



Aguda



Causas

Paracetamol



Não Paracetamol

Vírus

Auto imune

Drogas / DILI

Wilson

Budd Chiari

Gestação

**Causa não
identificada
(>30%)**



Dan L. Longo, M.D., *Editor*

Drug-Induced Liver Injury — Types and Phenotypes

Jay H. Hoofnagle, M.D., and Einar S. Björnsson, M.D.

Table 2. Phenotypes of Drug-Induced Liver Injury.*					
Phenotype	Type of Liver Injury	Latency	Enzyme Pattern	Typical Agents	Comments
Acute hepatic necrosis	Direct	Days	Marked, abrupt ALT elevations; mild Alk P and bilirubin elevations	Acetaminophen, aspirin, niacin, "Ecstasy"	Often due to overdose
Enzyme elevations	Direct	Days to months	Mild-to-moderate ALT or Alk P elevations	Many agents	Usually transient and asymptomatic
Acute hepatitis	Idiosyncratic, indirect	Days to months	High ALT elevations, modest Alk P elevations	Isoniazid, diclofenac	High death rate
Cholestatic hepatitis	Idiosyncratic	Weeks to months	High Alk P elevations, modest ALT elevations	Amoxicillin–clavulanate, cefazolin	Pruritus, early and prominent
Mixed hepatitis	Idiosyncratic	Days to months	Moderate ALT and Alk P elevations	TMP-SMZ, phenytoin	Usually benign, self-limited
Chronic hepatitis	Idiosyncratic, indirect	Months to years	Moderate ALT elevations with bilirubin elevations	Diclofenac, nitrofurantoin, minocycline	Insidious onset; may require glucocorticoids
Bland cholestasis	Unknown, possibly idiosyncratic	Months	Moderate ALT elevations, mild Alk P elevations	Anabolic steroids, estrogens	Pruritus, prominent and prolonged
Acute fatty liver, lactic acidosis, and hepatic failure	Direct	Days to months	Lactic acidosis, modest ALT elevations, hepatic failure	Stavudine, linezolid, aspirin (Reye's syndrome)	Mitochondrial failure, pancreatitis
Nonalcoholic fatty liver	Indirect, direct	Months	Mild ALT and Alk P elevations	Glucocorticoids, tamoxifen, haloperidol	Asymptomatic; fatty liver seen on ultrasound
Sinusoidal obstruction syndrome	Direct	Weeks	Variable enzyme elevations	Cancer agents, busulfan, gemtuzumab	Hepatomegaly, weight gain, edema, ascites
Nodular regenerative hyperplasia	Direct	Years	Minimal ALT and Alk P elevations	Thioguanine, azathioprine, oxaliplatin	Noncirrhotic portal hypertension



Tratamentos específicos

Paracetamol

N acetilcisteína precoce

Em até 10h de preferência.
Max. 24 h da ingesta

**Outras
drogas**

N acetilcisteína já
estudada, sem evidência

Auto imune

Corticóide

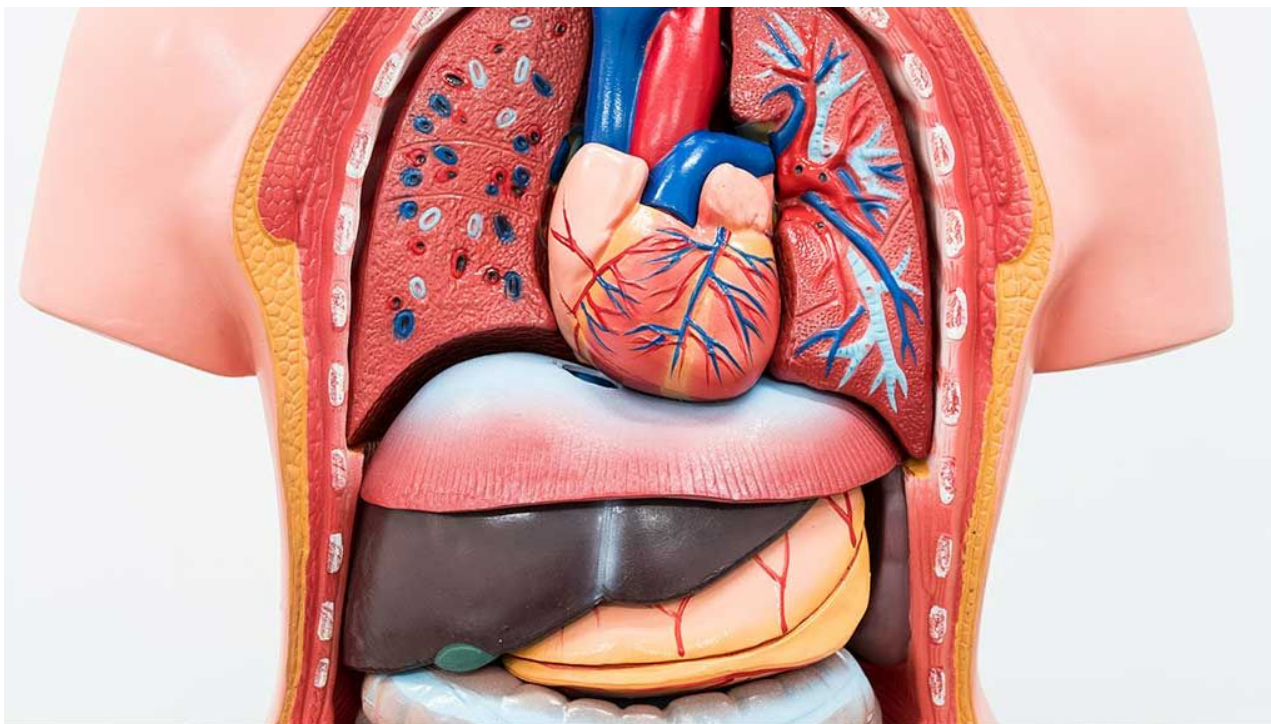
Gestação

Interromper



Ao que interessa na UTI ...

IHA é uma causa de disfunção de múltiplos órgãos



Objetivo do cuidado : suporte !



Por sistemas...

Neuro -

Resp -

Hemod -

Infecto -

Hemato -

Renal -

Profilaxias -

Dispositivos -



Por sistemas...

Neuro -

Resp -

Hemod -

Infecto -

Hemato -

Renal -

Profilaxias -

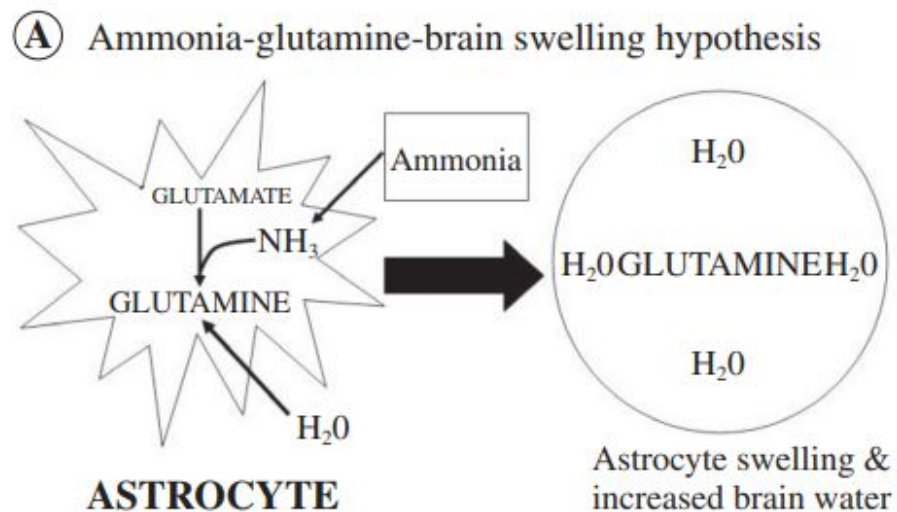
Dispositivos -



Encefalopatia Hepática

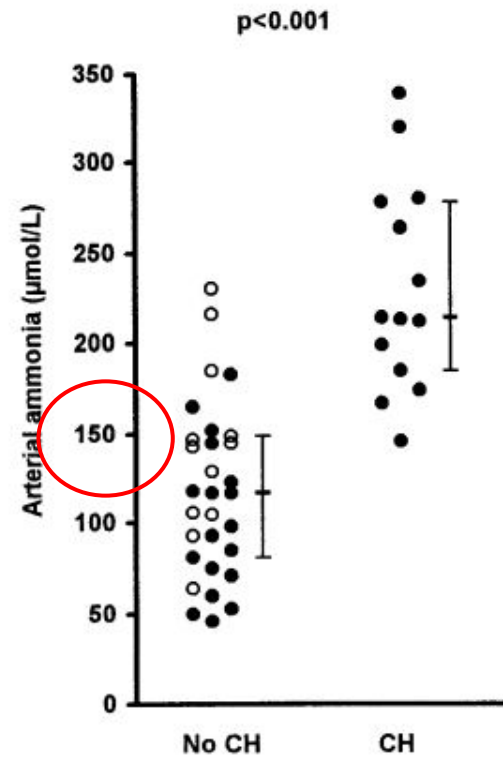
Preocupações:

Hipertensão intracraniana é uma causa importante de morte nesses pacientes;



Cerebral Herniation in Patients With Acute Liver Failure Is Correlated With Arterial Ammonia Concentration

JENS OTTO CLEMMESSEN, FIN STOLZE LARSEN, JENS KONDRUP, BENT ADEL HANSEN, AND PETER OTT



Como diagnosticar HIC ?

- Rebaixamento / coma
- Alterações pupilares
- USG bainha de nervo óptico
- DTC
- Tc de crânio



E a PIC ?

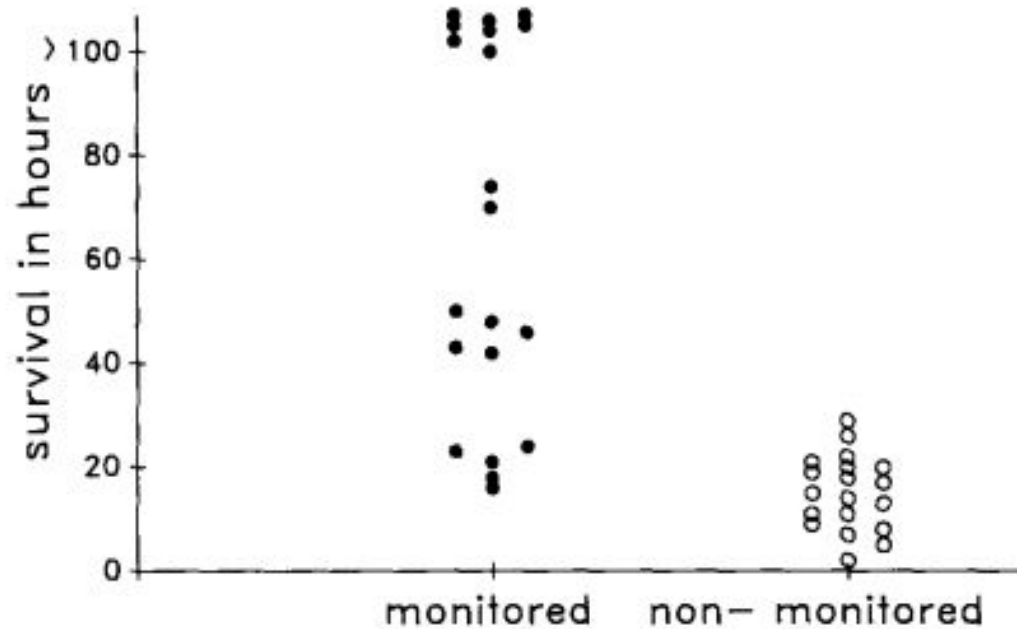
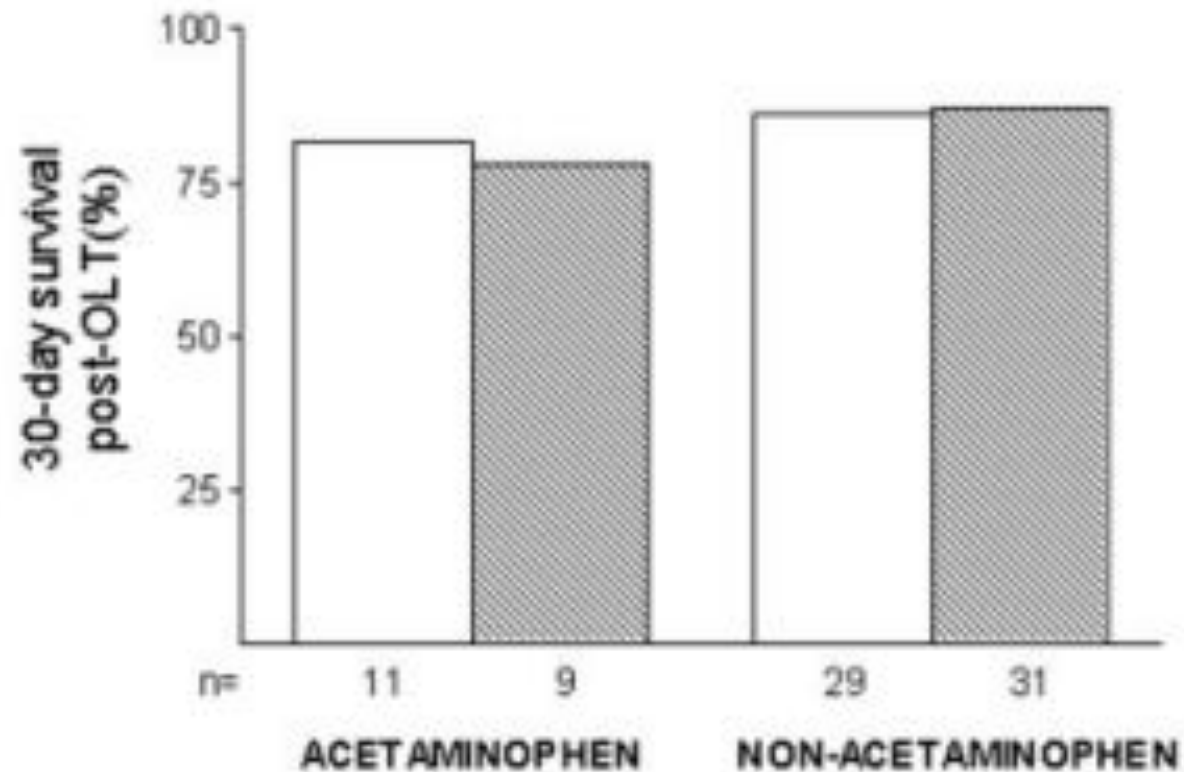


Fig. 1. Duration of survival from entry into grade IV coma to death in the monitored and non-monitored groups (for patients surviving > 10 h, $P < 0.01$).





30-day survival post-LT was similar regardless of ICP monitoring in the global series



Como tratar a hiperamonemia ?

Hemodiálise

- Uréia e amônia têm cinética semelhante em termos de clearance por difusão e/ou convecção.
- Ambas têm peso molecular baixo e volume de distribuição alto.
- Dose e intensidade da diálise são relevantes para manter amônia em níveis normais de maneira rápida.



Continuous renal replacement therapy is associated with reduced serum ammonia levels and mortality in acute liver failure

Filipe S. Cardoso, Michelle Gottfried, Shannan Tujios, Jody C. Olson, Constantine J. Karvellas ✉ For the US Acute Liver Failure Study Group,

-CRRT use on D1 → lower 21-day transplant-free mortality. (95% CI – 0,26-0,82)

-IRRT use on D1 → higher 21-day transplant-free mortality. (95% CI – 1,04-2,72)

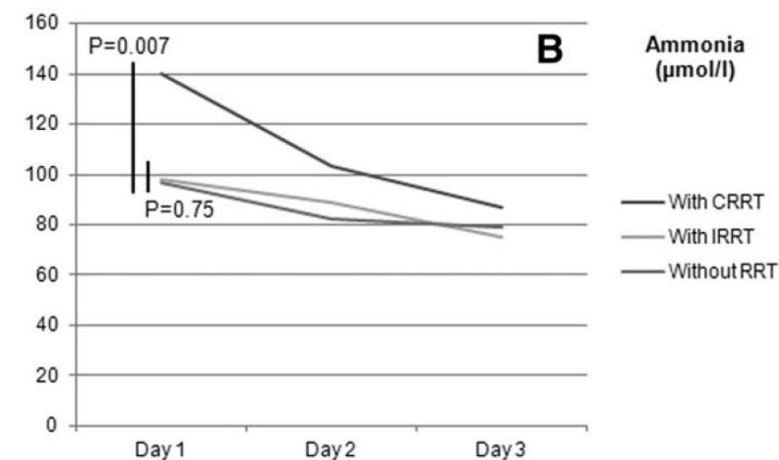


FIG. 1. (A,B) Renal replacement therapy on days 1 and 2 and median serum ammonia level (μmol/L) between days 1 and 3 post study admission (n = 340).



CIRRHOSIS AND LIVER FAILURE

Ammonia clearance with haemofiltration in adults with liver disease

Andrew J. Slack, Georg Auzinger, Chris Willars, Tracy Dew, Rebecca Musto, Daniel Corsilli, Roy Sherwood, Julia A. Wendon and William Bernal

Institute of Liver Studies, King's College Hospital Foundation Trust, London, UK

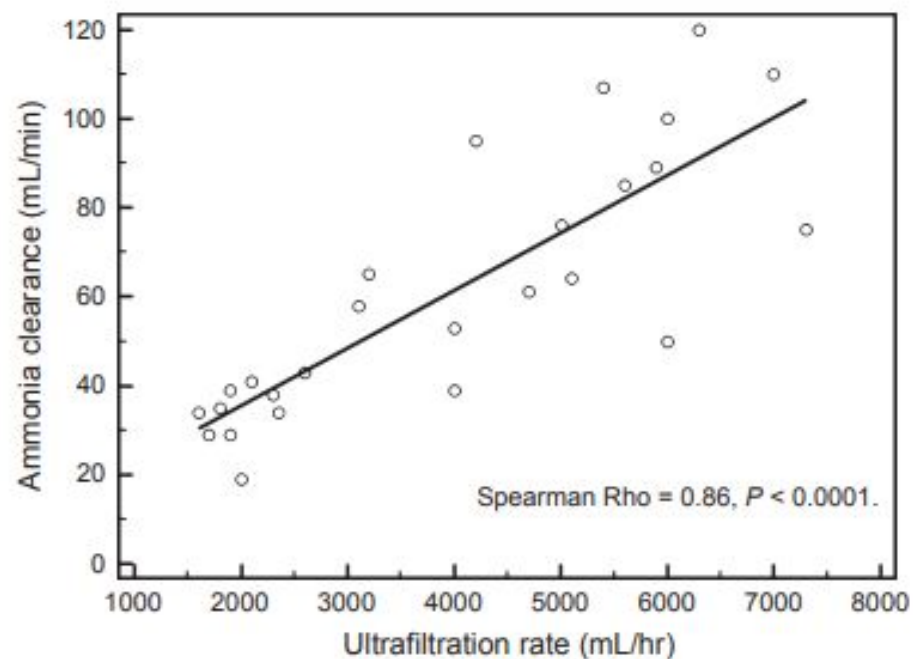


Fig. 2. Correlation between ultrafiltration rate and ammonia clearance after 1 h of continuous renal replacement therapy.



Peculiaridades da prescrição

- Sempre contínua
- Sódio alto
- Doses mais altas (?)

Anticoagulação ?

Citrato !!!

Lembrar que mesmo nestes casos o citrato pode ser usado e a intoxicação parece ser apenas um marcador de prognóstico.

Vigilância : hipocalcemia, acidose, lactato...



- Early institution of extracorporeal support (RRT) should be considered for persistent hyperammonaemia, control of hyponatraemia and other metabolic abnormalities, fluid balance and potentially temperature control (**evidence level III, grade of recommendation 1**).
- Anticoagulation of RRT circuits remain a matter of debate, and close monitoring of metabolic status should be undertaken if citrate is utilised (**evidence level II-2, grade of recommendation 1**).
- Continuous RRT should always be undertaken in the critically ill patient with ALF as opposed to intermittent haemodialysis (**evidence level III, grade of recommendation 1**).

Clinical Practice Guidelines



EASL Clinical Practical Guidelines on the management of acute (fulminant) liver failure[☆]

European Association for the Study of the Liver^{*}



Demais recomendações :

- Propofol é a droga de escolha para sedação;
- Sem necessidade de profilaxia para crises convulsivas;

Lactulose/Rifaximina não tem evidência em IHA e aumenta risco de distensão abdominal.



Por sistemas...

Neuro -

Resp -

Hemod -

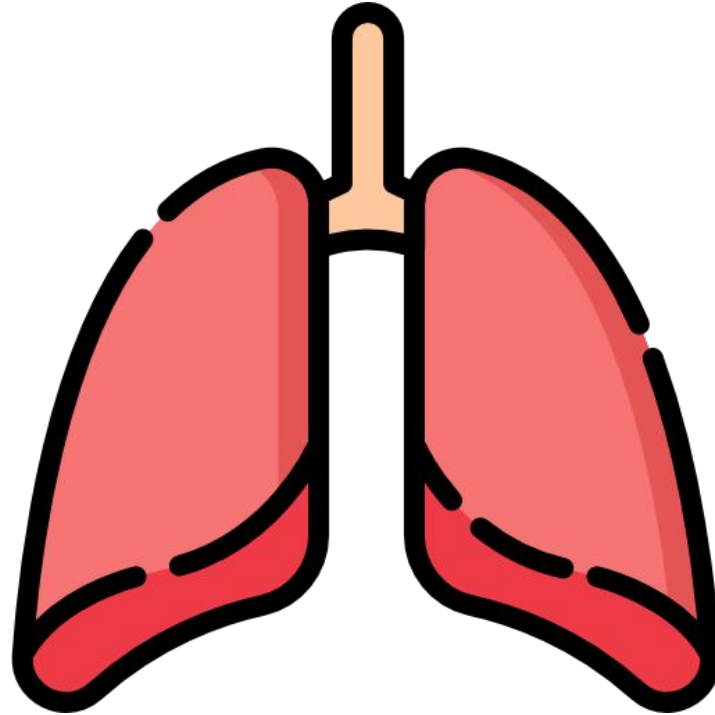
Infecto -

Hemato -

Renal -

Profilaxias -

Dispositivos -



Disfunção respiratória

- Paciente geralmente já está entubado pela EH;
- Pode evoluir com ARDS;
- Recomendações clássicas para VM.



Por sistemas...

Neuro -

Resp -

Hemod -

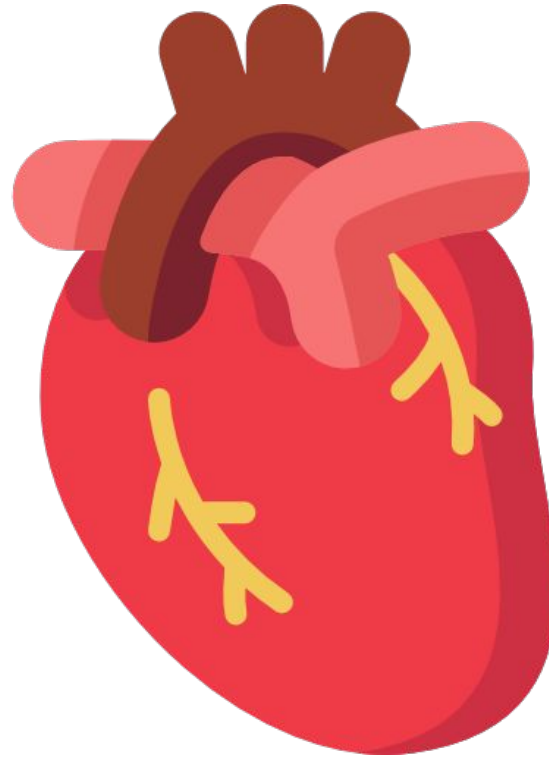
Infecto -

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Dispositivos -



Disfunção cardiovascular - choque hiperdinâmico

Síndrome Vasoplégica !!!

- Vasodilatação - Noradrenalina +/- Vasopressina
- Volume para tirar componente de desidratação / hipovolemia;
- Corticóide ? - Disfunção adrenal é muito prevalente;
- Lactato muito alto -> fígado não clareia / não é marcador ideal de hipoperfusão tecidual;
- Muita acidose = perpetua a vasoplegia.



Por sistemas...

Neuro -

Resp -

Hemod -

Abd/Nutricional -

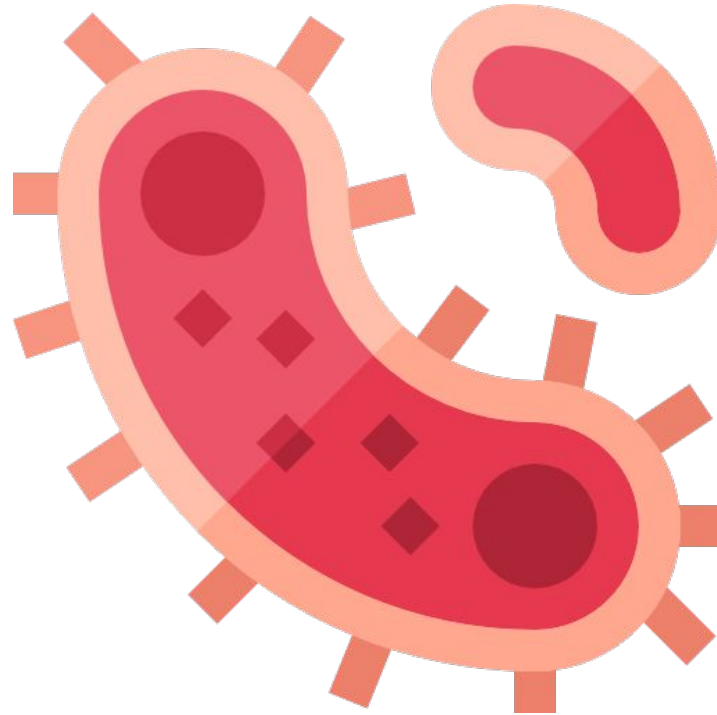
Infecto -

Hemato -

Renal -

Profilaxias -

Dispositivos -



Risco de infecção sobreposta

- Infecções por bactérias ou fungos são extremamente prevalentes nesta população;
- Sinais “clássicos” como febre e leucocitose estão ausentes em 30% dos casos;
- Incidência de infecções nos trabalhos variam de 35-80%;

Cada serviço tem sua recomendação.
Sugestão: Cefotaxima e Fluconazol.



Antibiótico profilático

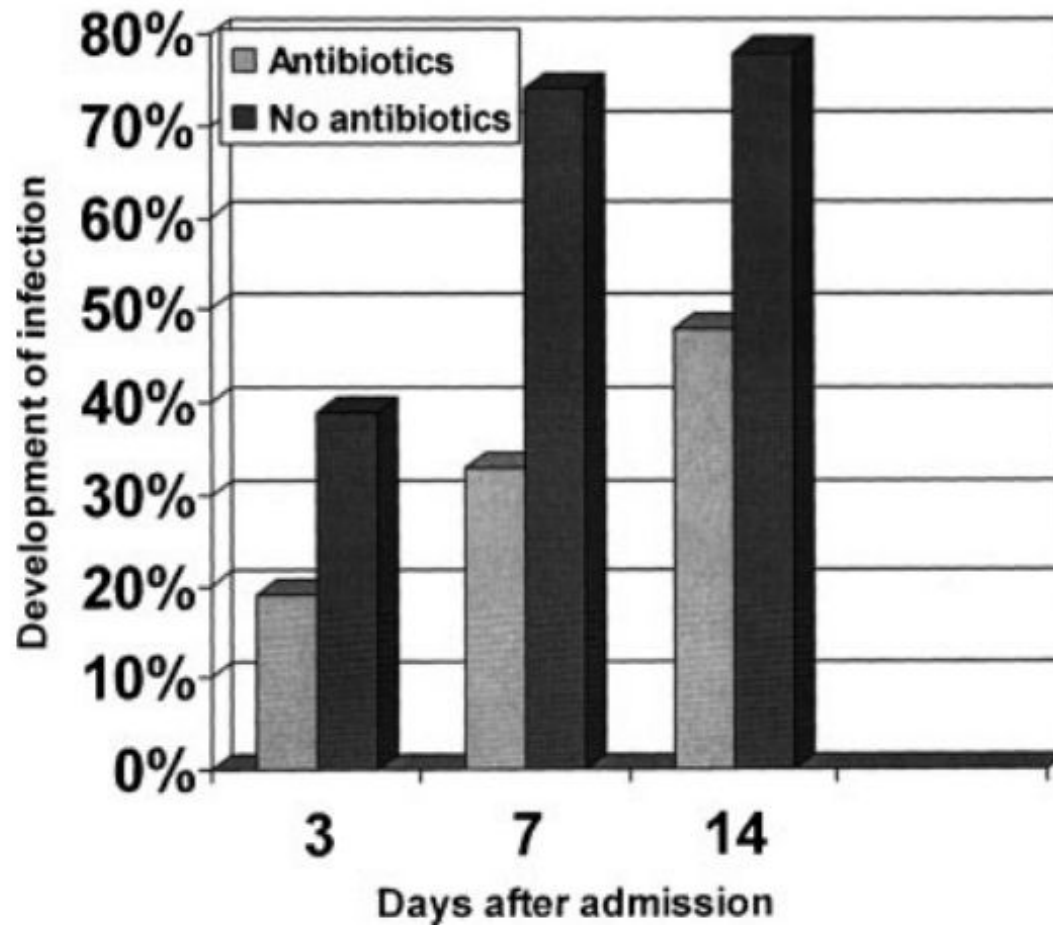


Figure 1 The effect of antibiotic prophylaxis on the prevalence of documented infection patients with acute liver failure. (Adapted from Salmeron et al.⁴⁷)

Sem redução
de
mortalidade



- Prophylactic antibiotics, non-absorbable antibiotics, and antifungal have not been shown to improve survival in ALF (**evidence level II-2, grade of recommendation 1**).
- Regular periodic surveillance cultures should be performed in all patients with ALF (**evidence level III, grade of recommendation 1**).
- Early anti-infection treatments should be introduced upon appearance of progression of hepatic encephalopathy, clinical signs of infections, or elements of SIRS (**evidence level II-3, grade of recommendation 1**).
- Antifungal therapy in those with prolonged critical care support for multiple organ failure should be considered, as guided by the use of biomarkers (**evidence level II-3, grade of recommendation 1**).



Por sistemas...

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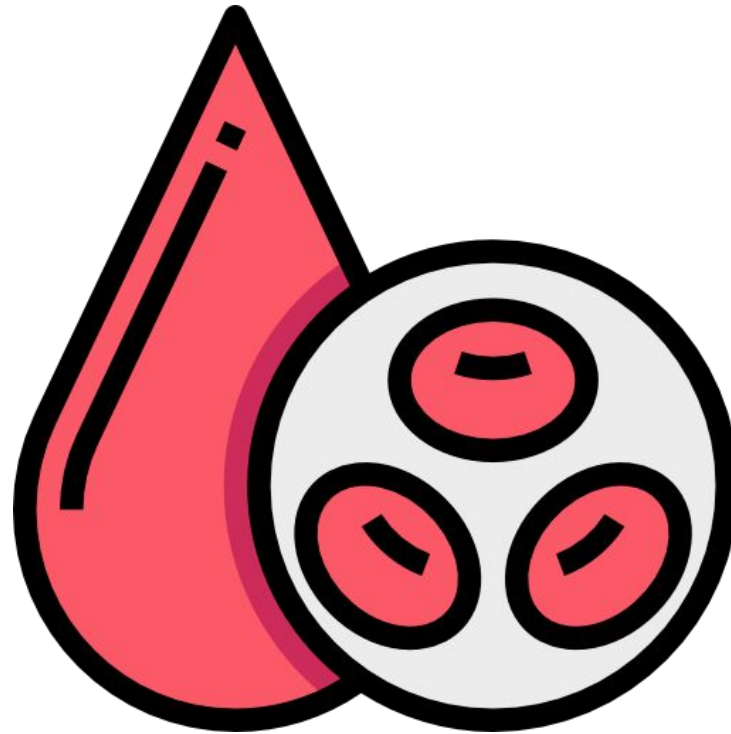
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Dispositivos -



Coagulopatia

- Guiar por tromboelastograma !!!!
- Muitas vezes paciente com INR alargado tem trombo normal;
- Corrigir INR pode atrapalhar critérios de transplantabilidade.

Alguns alvos sugeridos por guidelines :

**Hb > 7
Plaqueta > 50.000
Fibrinogênio 150-200**



Por sistemas...

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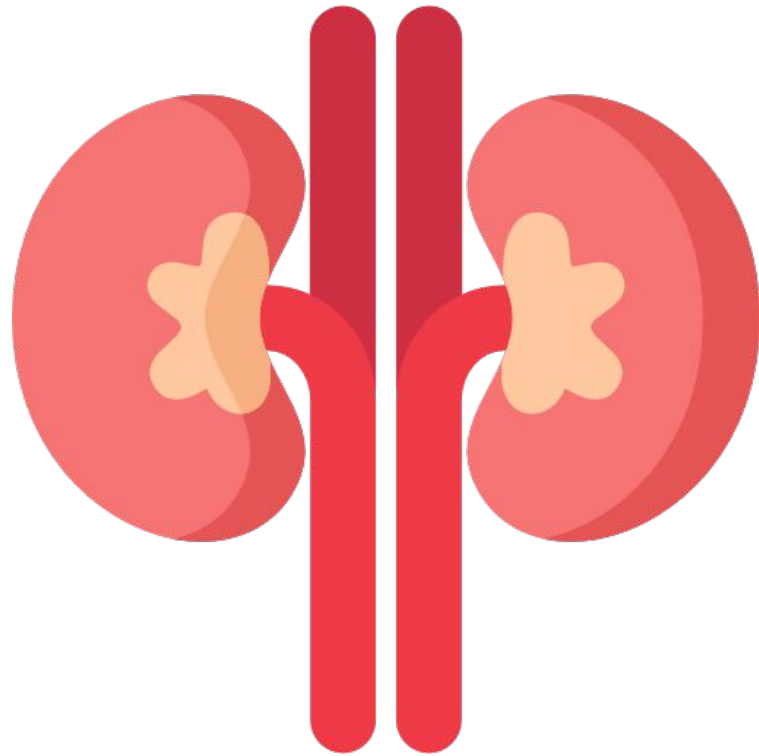
Infecto -

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Renal -

Profilaxias -

Dispositivos -



IRA

- 40-80% tem IRA associada IHA;
- Geralmente são dialisados precocemente pelo EH / hiperamonemia, antes da lesão renal propriamente;
- Hipofosfatemia parece ter relação com bom prognóstico por estar associado a regeneração hepática.



Por sistemas...

Neuro -

Resp -

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Resumindo:

IBP

TEV = não farmacológica



Por sistemas...

Neuro -

Resp -

Hemod -

Abd/Nutricional -

Infecto -

Hemato -

Renal -

Profilaxias -

Dispositivos -



TODOS



Terapias avançadas

ECAD (Extracorporeal albumin dialysis) : MARS e SPAD

A idéia é remoção de toxinas endógenas acumuladas na IHA através de um dialisato rico em albumina. Reduz níveis de bilirrubinas, melhora hemodinâmica e EH. Não reduz mortalidade.



Terapias avançadas

Plasmaférese

Research Article



EASL JOURNAL OF HEPATOLOGY

High-volume plasma exchange in patients with acute liver failure: An open randomised controlled trial

Fin Stolze Larsen^{1,*}, Lars Ebbe Schmidt¹, Christine Bernsmeier², Allan Rasmussen³, Helena Isoniemi⁴, Vishal C. Patel², Evangelos Triantafyllou², William Bernal², Georg Auzinger², Debbie Shawcross², Martin Eefsen¹, Peter Nissen Bjerring¹, Jens Otto Clemmesen¹, Krister Hockerstedt⁴, Hans-Jørgen Frederiksen⁵, Bent Adel Hansen¹, Charalambos G. Antoniades^{2,6,†}, Julia Wendon^{2,†}

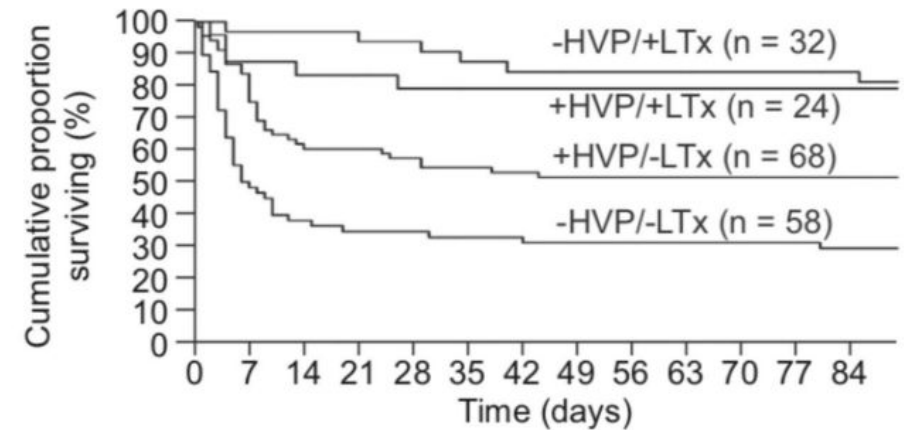


Fig. 2. Survival in the groups, in the two groups receiving SMT (standard medical treated group) with and without emergency transplantation (–HVP +LTx vs. +HVP–LTx) and the two group receiving SMT with and without emergency transplantation (–HVP–LTx vs. +HVP–LTx) (LogRank: $p = 0.0058$) and Cox proportional hazard: LTx: $p < 0.0001$; HVP: $p = 0.0076$).



Terapias avançadas

Research Article



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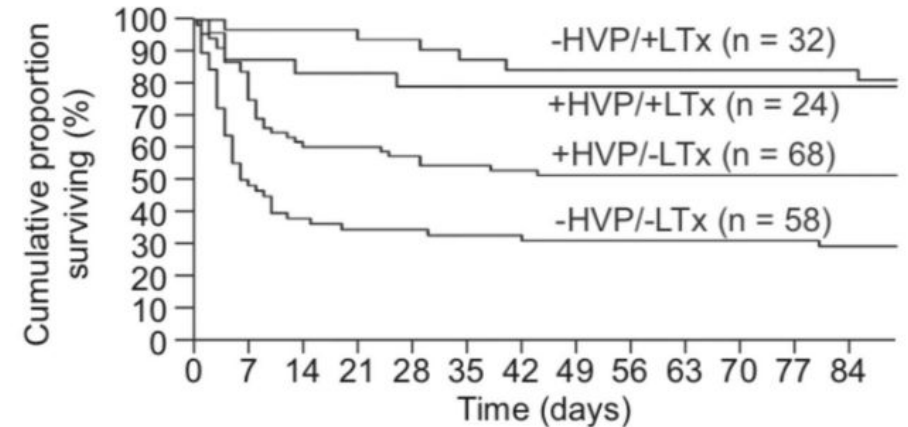


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58% paracetamol

15% DO PESO IDEAL

(8-12L PFC/DIA, 3 DIAS)



Transplante Hepático

King's College criteria

ALF due to paracetamol

- Arterial pH <7.3 after resuscitation and >24 h since ingestion
- Lactate >3 mmol/L or
- The 3 following criteria:
 - Hepatic encephalopathy $>$ grade 3
 - Serum creatinine >300 $\mu\text{mol/L}$
 - INR >6.5

ALF not due to paracetamol

- INR >6.5 or
- 3 out of 5 following criteria:
 - Aetiology: indeterminate aetiology hepatitis, drug-induced hepatitis
 - Age <10 years or >40 years
 - Interval jaundice-encephalopathy >7 days
 - Bilirubin >300 $\mu\text{mol/L}$
 - INR >3.5

Beaujon-Paul Brousse criteria (Clichy)

- Confusion or coma (hepatic encephalopathy stage 3 or 4)
- Factor V $<20\%$ of normal if age <30 year
or
- Factor V $<30\%$ if age >30 year



Conclusão

Reconhecer precoce !!! (EH < 6 m >> IHA)

Piora EH (sonolência) >> IOT + Depuração amônia !

Não corrigir o INR se não tiver sangramento !

Retirar causador !

Avaliar Transplante !!!!



Obrigada

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